China and the Global South: A Gravity Model Analysis to Study the Factors Affecting Trade Activities Between China and the Southern Countries*

Aru Bhardwaj
Research Intern,
Jindal Centre for the Global South
Email: arubhardwaj@hotmail.com

August 2020

Abstract

China accounts for more than 70 percent of market gains in south-south trade. Since the accession to the World Trade Organization (WTO), in 2001, China's trade activities, motivated by the needs of raw materials for industry and oil, with the southern countries have been widened. The largest part in the bilateral trade, between China and the global south, is of exports from China to these countries, mainly consists of labor-intensive goods. This paper studies the trade activities, and effects of geographical disadvantages, exchange rates, population, and sizes of the economies, between China and the southern countries. It applies the gravity models on the panel data from 93 countries from 2001 to 2019. Apart from the gravity models, it also applies the traditional technique of Ordinary Least Squares (OLS) to draw a comparison. The results show a strong adverse effects of population growth of China on the bilateral trade activities and also shows the negative coefficient values for the landlocked countries. The paper, however, finds no effects of exchange rates on the bilateral trade activities, as interpreted otherwise by some researches. The economic size of both, China and the trade partners, have a positive coefficient value, showing that the economic mass (or size of the economy) has a favourable relationship with the growth of bilateral trade.

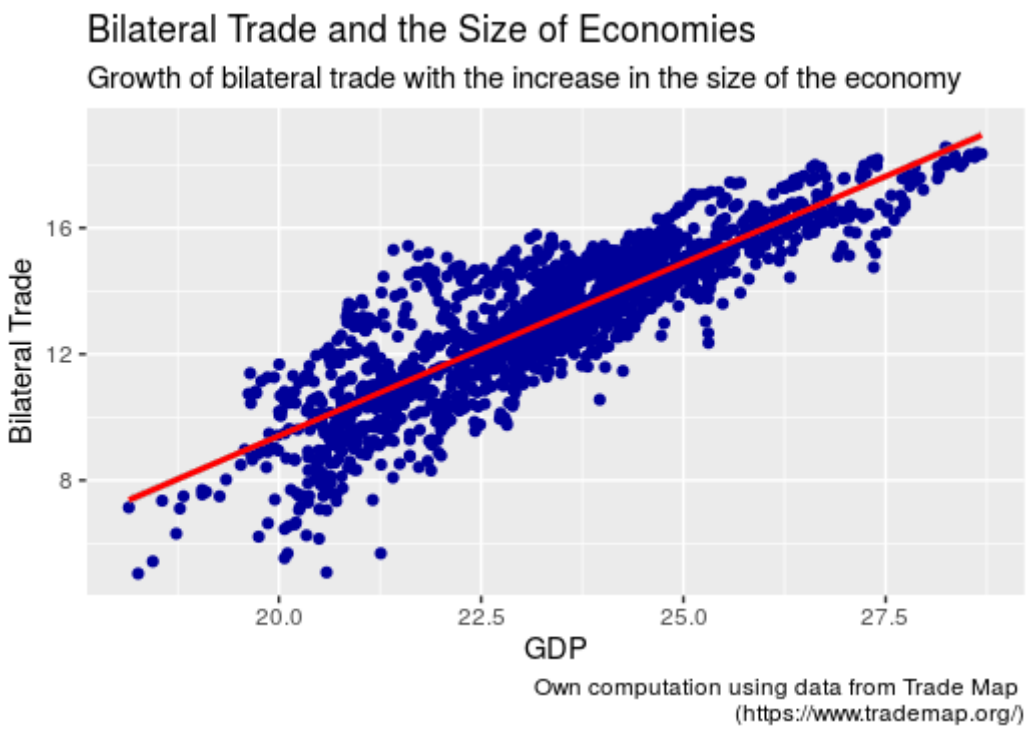
*This is an independent research project undertaken during the Internship at the Jindal Centre for the Global South. I am thankful to JCGS for organizing multiple webinars discussing the development, trade relations and political economy of the region, which helped me a lot in developing an idea to study the trade activities between the China and the southern countries. All errors are mine.

Introduction

China introduced economic reforms in the 1970s and opened its market to connect with the rest of the world. After China, other countries in the region embarked on the liberal policies and proceeded towards the trade liberalization. Later, in the 1990s, the globalization movement gave a new momentum to the market-oriented policies, which helped all those countries in integrating with the world economy and resulted in an overall increase in the income levels and economic growth (Sakyi et al., 2015) Benefits of globalization were enjoyed by all in the longer period, however, some industries in the countries were affected tremendously due to the foreign competition, too. Since then, China's trade interaction with the developing countries keep on expanding and today, China plays a lead role in south-south trade cooperation. Aksoy and Ng (2013) argues, about 70 percent of market gains from south-south trade belongs to Peoples' Republic of China (PRC) only. The major factor which jeopardizes the south-south trade cooperation among the countries are the resource constraints. Since the accession to the World Trade Organization[1], China emerged as a major exporter in the global trade, especially in the labor-intensive market. Its accession to the World Trade Organization (WTO), in 2001, was a milestone reform of openness and integrating with the world economy and, on that event, the WTO's Director-General said *"International economic cooperation has brought about this defining moment in the history of the multilateral trading system. With China's membership, the WTO will take a major step towards becoming a truly world organization. The near-universal acceptance of its rules-based system will serve a pivotal role in underpinning global economic cooperation"*.

The south-south trade is mainly motivated by PRC's need for raw materials for its domestic industries and for fulfilling this need it is engaging with the small countries, including island nations, in Africa and Latin America (Ferchen 2011; Alden et al. 2008). Being a dominating player in south-south trade, studying the role (and the trade activities) of China with the global south becomes essential. In almost two decades, China has emerged as a major global southern power and has become another finest example of super economic power dominating the global economy. President Xi Jinping, in 2013, right after assuming the office of Peoples' Republic of China (PRC), started a flagship program named as Built Road Initiative (BRI). After the financial crisis of 2007, economies like India and China took a lead in becoming the southern powers which shifted the market power from the global north to the global south. Researchers, such as Zhou and Haibing (2019), view the BRI as an opportunity of re-energizing the south-south trade cooperation. At the beginning of the year 2020, which marks the seventh anniversary of the BRI project, the Chinese government is looking towards encour-

[1] On September 17, 2001, the World Trade Organization (WTO) completed all the negotiations on entry of China.



Notes: This plot shows the relationship between the bilateral trade, between China and countries of the global south, and size of economies of trade partners. The data is in logarithmic form and shows the positive trend between the two variables, with $R^2 = 0.76$

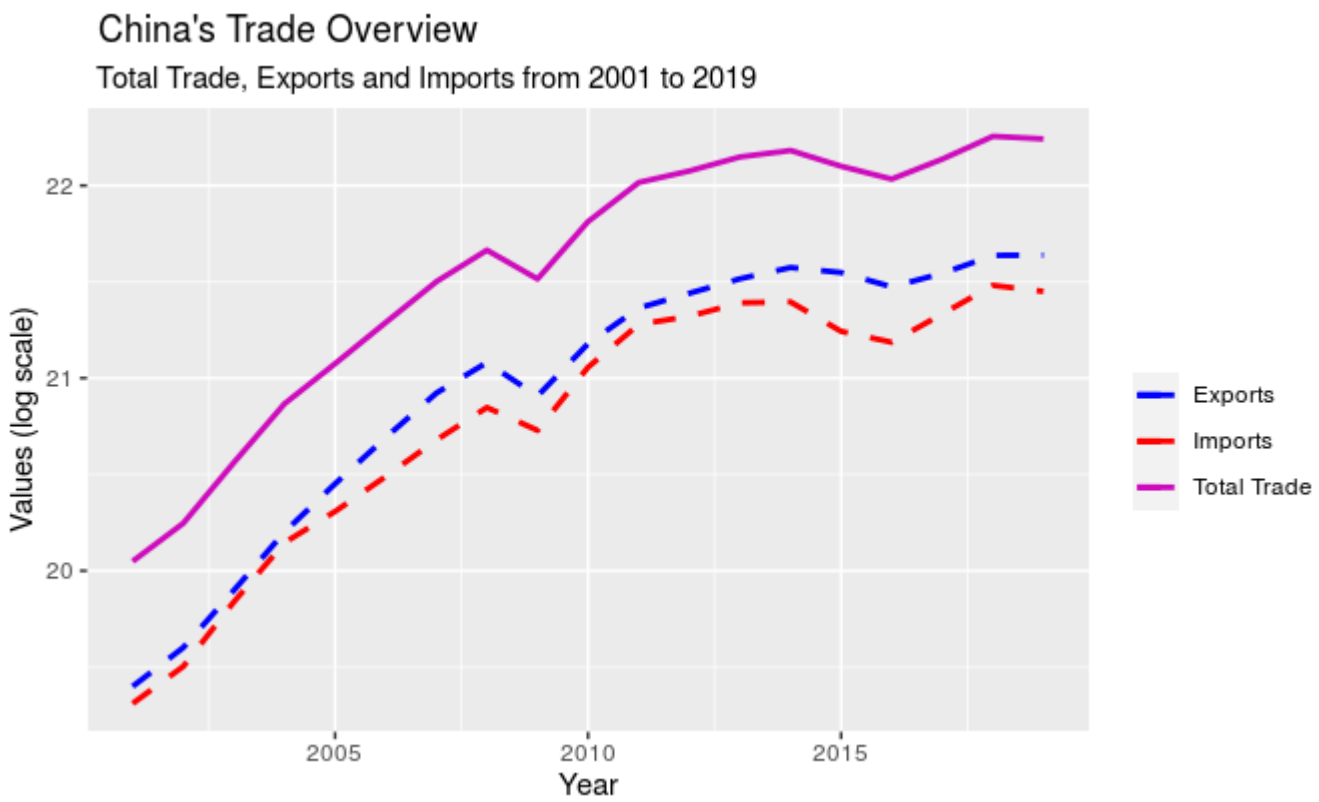
Figure 1: Plot showing the relation between GDP versus Bilateral Trade Flows

aging imports from the developing countries promising the transparency, open markets and a focus on the quality[2]. The obvious reason for this shift is the need for more raw materials for expanding the industrial operations and calming down the anti-china resistance stirring in the developing countries.

Over the years, both the bilateral trade flows between China and its trade partners in the developing world have been increased astonishingly. Since most countries are following a less restrictive approach to trade and opening up following the policies of liberalization. The globalization movement of the 1990s challenged the trade control practices, followed the countries for protecting the domestic industries, and established a liberal economic world order. This, later in 1995, gave birth to the successor body of the General Agreement on Tariffs and Trade (GATT), the World Trade Organization (WTO). The entry of China in the WTO required the administration to undertook necessary changes in its economy. However, it has always been accused of "cheating the system in various ways" and does not fully endorse a market-oriented approach (Bacchus et al., 2018).

Today, China's dominance on the world trade reflects a success story of a developing country, through attaining the benefits of globalization and less restrictive trade practices. The data shows that,

[2]see: Joshi, M. (2020) *With BRI 2.0, Xi Jinping pledges to step up China's game*, Commentary, Observers Research Foundation. Available at: link (Last accessed: 8 August, 2020)



Data Source: UN Comtrade and Various Sources, retrieved form Trade Map

Figure 2: Overview of China's Trade.

over the years, there has been a strong positive relationship between the trade and size of economies. As shown in the figure 1, from 2001 to 2019, both the bilateral trade flows and GDP size have grown together. Larger size of economy implies bolstered industrial operations which reflects the development of trade over the period. Economic size is a common parameter which also affects the trade potential of the countries. Past studies have shown that distance and economic size are the *two* strong determinants that affect the flow of trade from countries. In the case of China, too, as discussed later in the paper, economic size has a strong effect over trade, however, results here refuted the claim that distance (proxy for transportation costs) between the countries has any negative consequence on bilateral trade flows. Since, this paper finds no such effect of distance on the bilateral trade between China and the global south.

China since 2001, always have an overall trade surplus: overall exports from China always exceeds its overall imports. Figure 2 portrays the picture of China's overall trade effects during the years of financial crisis, too, as the curve gets bend in that period In 2015, China achieved the highest trade surplus of $594 billion, but since then the year-on-year growth of surplus is declining, which is because of the so-called Asian trade slowdown (Hong et al., 2017).

Country	**Trade (USD Mil.)**	**Total Share (%)**
Top Importers		
United States	2,314,963	12.49
China	1,680,279	9.07
Germany	1,183,472	3.56
Hong Kong, China	659,846	3.56
United Kingdom	650,901	3.51
Top Exporters		
China	2,750,811	14.57
United States	1,573,643	8.34
Germany	1,436,856	7.61
Japan	807,965	4.28
Korea, Rep.	601,732	3.19

Source: World Trade Summary 2018 Data. WITS, World Bank

Table 1: Top 5 Importers and Exporters..

Imports by the PRC accounts for about 9.07 percent of the world's share, and it has the highest exports in the world, followed by the United States (8.34 percent), with a share of 14.57 percent of world's total exports. As shown in table 1. It is the only country with a double-digit share in the exports of the entire world. Interestingly, it is exporting about double of what the US exports are. The dominance of China in the world market is through its trade activities of importing oil and raw materials, and exporting the labor-intensive finished goods. The top imports include: natural resources, machinery, plastics, precious metals, copper and chemicals. Whereas, the top exports from China includes mobile phones, computers, integrated circuits, processed petroleum oil, appliances, and automobiles. Most exports, here, includes the processed commodities on which values are added by the industries in China.

This paper estimates the impacts of hindrances due to geographic limitations, exchange rates, distance between the countries, population, and size of the economy, using the gravity models. It is divided in four sections, including the introduction. The next section will discuss the trade relations between the China and southern countries, which will include a brief discussion on the BRI's role in strengthening the Chinese influence in these countries. Then, in section *three*, gravity models are interpreted and discussed. In the same section results from both OLS estimates and gravity models under different settings are interpreted. The last section consists of conclusion of the paper.

Trade Relations Between China and the Global South

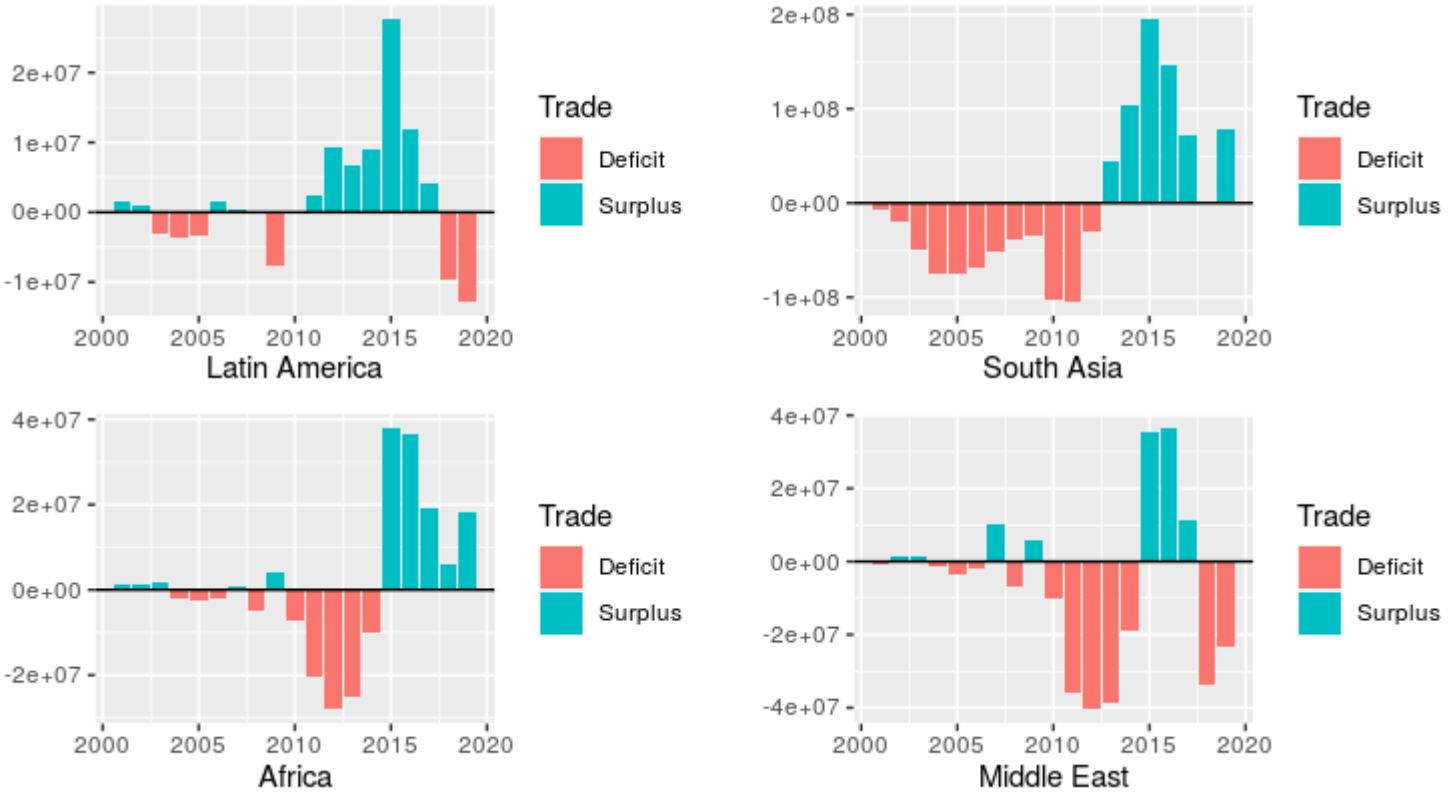
Diplomatic relations of China with the southern countries are motivated by the need for development in the global south. Through BRI, Chinese involvement in the political process of the countries enlarged and strengthened, since, by involving countries in the BRI project, China lends them money over and beyond their capacity of repayment. It was termed as a "debt-trap diplomacy" by Chellaney (2017). For instance, the Chinese administration gave all the assistance asked by Sri Lanka for development of the port and in the end, China claimed all rights over it[3]. It was claimed that the BRI will ease the trade for landlocked countries like Afghanistan, Kazakhstan, Kyrgyzstan, Tajikistan, and Turkmenistan. The road will connect them directly to Beijing, and the Europe on other end. It could energize the bilateral trade ties between the nations and the Beijing but will give a chance to the 'rent-seeking' behavior of China to emerge withing those connected countries and increase the influence of state-owned and private Chinese firms.

It has reached to the even smaller countries of Africa and Latin America. In the past two decades, the bilateral trade between China and Latin America has grown at an astonishing rate, from $17 billion in 2002 to $315 billion in 2019. Since 2004, when the former Chinese President Hu Jintao visited the region and subsequently various leaders from the region visited China, the trade ties between the region and China strengthened. Last year, in 2019, President Xi Jinping also made a visit to four Latin American countries for strengthening the bilateral ties and discuss the prospects of BRI for the region[4]. During these years, both the exports and imports between the countries have grown spectacularly. Figure 3 shows that mostly China had a surplus from trade with Latin America and the Caribbean. Using the database of 620 goods Blázquez-Lidoy et al. (2006) developed two indexes to study the competition between the China and 34 economies, from which 15 were Latin American. The results conclude that the European and Asian counterparts face much stronger competition from China than the Latin American countries.

Figure 4 shows that China always had surplus trade balance, since joining the WTO, in 2001. The effects on China's trade during the financial crisis of 2007 and recent Asian trade slowdown are clearly significant in the figure. Despite all the speculation in the global economic environment, China's trade always remained in surplus.

[3]see: Abi-Habib, Maria. (2018): *How China Got Sri Lanka to Cough Up a Port*, June 25, 2018. New York Times. Available at: link

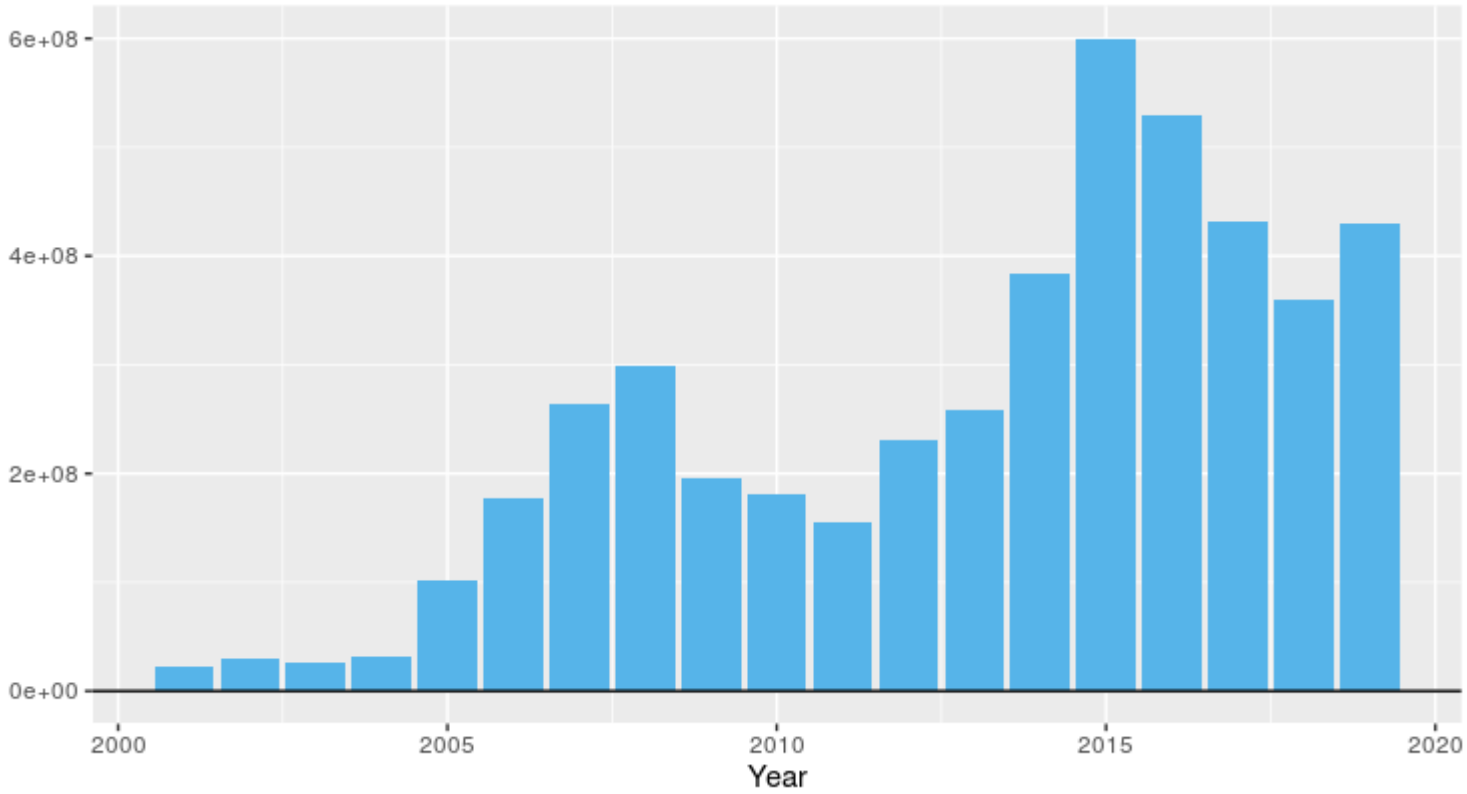
[4]see: Xinhua, (2019): *Feature: Xi spearheads closer China-LatAm cooperation for common prosperity*, November 12, 2019. Xinhua Net. Available at: link



Source: Own computation using data from Trade Map (https://www.trademap.org/)

Note: The figure shows the difference between exports and imports of China to Latin America, South Asia, Africa, and the Middle East, from 2001 to 2019. The reddish bars represents the negative net exports, which means China had a deficit within the trade in the region, in that specific year. In the South Asia panel, it clearly shows the surplus for China since 2013, the same results are reflected in the Africa panel. In Latin America, except for 2018 and 2019, China had a surplus since 2011.

Figure 3: Net Exports from China (Region wise)



Source: Own computation using data from Trade Map

Note: The figure shows the difference between exports and imports of China. It portrays an interesting picture of China, since joining the WTO, China never had any trade deficit overall. After becoming a member of the WTO, China has shown a significant growth in trade with the developing countries, where it imports the raw material and exports the manufactured goods to them.

Figure 4: Net Exports from China to the World

Jenkins et al. (2008) argues that there are two impacts of their engagement in the Latin America: growth impact and poverty impact. The effect of adverse trade balance for Latin America affects the growth in the region in the short-run, and in the long-run, it is leading to deindustrialization in the region because of deteriorated effects on the technological development. According to Jenkins and Edwards (2004) some countries in Asia and Latin America are successful in exporting more to China but most of their exports include "non-labor-intensive agricultural products and extractive products", which do not have a significant positive impact in improving the living standards for the poor, since the production of these goods are not directly associated with them. Asian countries have a more significant amount of imports from China in comparison to Africa and Latin America, therefore the chances of facing competition in the *third market* are more in the Asian countries than the other regions.

Despite the revolutionary progress made by China, it cannot claimed to become an autonomous engine of economic growth in Asia. The presence of emerging markets like Thailand, Philippines, India, Singapore, Indonesia, Korea, Malaysia, Pakistan, and Taiwan have much broader significance in the region. However, China has weakened the position of economies like the USA and Japan in the Asian trade (Gaulier et al., 2007). Table 1 showed that among the top five exporters in the world, three are located in Asia. Similarly, the emerging markets in Africa, such as Botswana, Ghana, Kenya, Mozambique, Nigeria, Tanzania, Uganda, and Zambia, are the major engine of growth in the region. In the past two decades, flows of African exports to China have also increased but according to the study by Hu and Marrewijk (2013), using the gravity model, trade with Africa has very limited prospects in strengthening the trade flows from Africa.

African economies exports raw materials for the industry to China, which includes the natural resources, too. For instance, Sudan exports 60 percent of its extracted oil to China which fulfills the 5 percent requirement of its industry. Apart from that, it also imports many commodities like mineral ores, petroleum, and agricultural products and major exports to Africa includes manufactured goods (Lammers, 2007). Not limited to this, large state-owned and private Chinese firms are heavily investing in the Sub-Saharan Africa (SSA) and extracting the natural resources of the region, which in turn is benefiting the SSA through aid, trade and foreign direct investment (Kaplinsky and Morris (2009); Kaplinsky and Morris (2010)). China's expanding influence through soft power[5] in South Africa led to the establishment number of Mandela Institutes in the country. Due to this influence in South Africa,

[5] *Soft power* is a term coined by the Harvard academic Joseph Nye, in his widely read book, titled *Soft Power: The Means To Success In World Politics* originally published in 2004. However, he first used the term two decades before the publication of this book, in the 1980s. The term refers to the influence over other countries through cultural linkages, linguistic similarities, economic help, and other non-hard persuasive approaches. Interestingly, this is achieved partly because of government but partly despite government.

Chinese influence all over Africa has strengthened symmetrically (Bodomo, 2009). However, some commentators argued that China is practicing "hard power" in Africa for dominating its strategic and economic interests, and taking control over the political process in the region, too. (Nantulya (2019); Sall (2016)). There is a good number of Chinese citizens working in the Chinese state-owned and private firms, based in Africa. They include some research groups, skilled and unskilled workers, and government officials, however, despite the increasing influx of Chinese workers in the region very little amount of research is addressing the migration patterns and its impact (Park, 2009). Kuang (2008) characterizes the Chinese migrants in Africa in *three* categories: labor migration, entrepreneurial migration, and proletarian migration.[6]

In Asia, China is leading the trade integration by connecting to Europe through its BRI project (Vines, 2018). The BRI initiative has strong prospects of changing the political economy of Asia and especially benefiting the Central Asia region in trade. The *new silk road* and *maritime silk road* initiatives, too, opened the doors of many opportunities for the Asia and the Middle East, and also proved beneficial for the poor countries in East Africa. Majorly, these initiatives of China affected the political economy of Euroasia. The landlocked countries in South Asia are going to be benefited by it, for instance, the China-Pakistan Economic Corridor (CPEC) is connecting the landlocked countries in the region. Countries like Nepal, Bhutan, and Afghanistan are majorly engaged in trade with its neighboring countries and have no or very little access to the other foreign markets (Bhatnagar and Ahmed, 2020).

[6]Labour migrants include people hired by the firms in the public works and entrepreneurial migrants are the migrants who are living in the region because of their business interests. According to Kuang (2008) these proletariat migrants are in transition to seek opportunities in Europe. Understandably, none of these migrants have any permanent ambition of being in the region, but are living for there for the specific interests.

The Model: Discussion and Interpretation

The use of gravity equations to determine the effects of factors on the bilateral trade between the countries is not a new affair. Gravity models have a history of more than 60 years in trade literature and are considered as "workhorse" models of international trade. This model, derived from physics in the 20th century, has enjoyed a great attention in Economics. Tinbergen (1962) was pioneer in deriving the gravity equation for trade from the Newton's law of gravity, and applying in his works on international trade patterns. Numerous researchers since then have contributed to the development of gravity equations and also developed different methods in bolstering the significance of method by extending its forms (such as, Anderson and Van Wincoop (2003); Silva and Tenreyro (2006); Anderson (1979) and many more). The standard Newtonist gravity equation is $F_g = G \times \frac{m_1\ m_2}{r^2}$, where m_1 and m_2 represents the mass of two different objects and r represents the distance separating the objects' centers, and G here is a constant. After modifying the Newtonist equation and reinterpreting its variables, the standard equation used in trade and migration models is the following:

$$X_{ijt} = A \times \frac{GDP_{it} \cdot GDP_{jt}}{d_{ij}^2} \tag{1}$$

where, in this case, X_{ijt} is the bilateral trade between i and j^{th} country in the tth year. The GDP_{it} and GDP_{jt} stands for the economic mass (size of economies) in the tth year (Krugman and Obstfeld, 2005). In our model, i^{th} country is China, which will remain the same throughout this paper, and j stands for its trade partners which in this data set are 93 in number—southern countries. Tinbergen (1962) version of gravity equations reflects that the trade between the two partners is stochastically determined. In the equation form it can be written as:

$$ln(X_{ijt}) = \alpha + \beta_1\ ln(GDP_{it}) + \beta_2\ ln(GDP_{jt}) + \tau\ ln(D_{ij}) + \gamma\ \theta_j + \epsilon_{ijt} \tag{2}$$

In equation 2, α is a constant term and θ_{ij} represents the policy or situation one is interested in studying. Distance between the two trade partners is represented by D_{ij} and ϵ_{ij} is an i.i.d stochastic term. In our analysis, this equation will take a form of

$$\begin{aligned} ln(X_{ijt}) = \alpha_i + \beta_1\ ln(GDP_{it}) + \beta_2\ ln(GDP_{jt}) + \tau\ ln(D_{ij}) + \kappa_1\ ln(Pop_{it}) \\ + \kappa_2\ ln(Pop_{jt}) + \gamma_1\ \theta_j + \gamma_2\ XE_{ijt} + \epsilon_{ijt} \end{aligned} \tag{3}$$

where θ_{ij} represents the dummy variable for the Landlocked country (1 = true and 0 = false), and XE represents the exchange rates between China and trade partners in the tth year. The exchange rate is calculated as:

$$XE_{ijt} = \frac{Average\ annual\ exchange\ rate\ in\ USD\ of\ China\ in\ t^{th}\ year}{Average\ annual\ exchange\ rate\ in\ USD\ of\ trade\ partner\ in\ t^{th}\ year} \tag{4}$$

Literature on gravity models supports the addition of exchange rate in the gravity model, since it helps to explain the variations in the bilateral trade among the participants (Bergstrand, 1985). Dell'Ariccia (1999) finds a small negative relationship between the exchange rates and bilateral volatility on trade. Tinbergen used the gravity equation to explain the patterns of trade that would prevail in the absence of any hindrance (De Benedictis and Taglioni, 2011). Even today, researchers are interested in understanding the benefits that occurred because of 'ease' in trade restrictions or the emergence of the Free Trade Areas (Kepaptsoglou et al., 2010). Studies such as Urata and Okabe (2010), have immensely contributed in explaining the impacts of free trade agreements (FTA) and discussing the trade creation and trade divergence effects with the help of these models. However, according to Baier and Bergstrand (2004), unlike in the "CGE models, the gravity equations do not explain the (endogenous) formation of FTAs". Evaluating the impact of free trade agreements on the bilateral trade flows is beyond the scope of this paper, since there are only 6 countries (out of 93) in our data set who have FTA with China[7].

Data

The panel data used here is scaped from multiple sources. All details related to description, sources, and measurement unit of variables are shown in table 2. Samples of 93 southern countries are scaped, for the period 2001 to 2019, which are engaged in trade with China. Distance between the national capitals of China and its trade partner is obtained from the search on Google Maps. Data on Gross Domestic Product (GDP) in current USD form is obtained from the World Bank national accounts and OCED national accounts data files, available of the World Bank's data repository.[8] Population data of countries is extracted from the United Nations' World Population Prospects. The dummy variable for landlocked countries is assigned with help of list available at Geography Realm.[9] In this sample over

[7]Only six countries in our data set of 93 countries have FTA with China, these are Chile, Mauritius, Maldives, Costa-Rica, Peru and Pakistan. The list of FTAs between China and the countries is available at: Ministry of Commerce's website

[8]Data repository of the World Bank: https://data.worldbank.org/

[9]The list of all landlocked countries can be accessed at: https://www.geographyrealm.com/landlocked-countries/ (last accessed: August 7, 2020.)

Variable	Description	Unit	Data Source
X_{ijt}	Bilateral trade between China (i) and the trade (j) partners during the year t	in USD '000	UN Comtrade
$distance_{ij}$	Distance between Beijing (i) and national capitals (j) of trade partners	Km	Google Maps
GDP_{it}	Gross Domestic Product of China in the tth year	Constant USD	World Bank and OCED
GDP_{jt}	Gross Domestic Product of trade partner in the tth year	Constant USD	World Bank and OCED
Pop_{it}	Population of China in the year t	units	United Nations
Pop_{jt}	Population of trade partner in the year t	units	United Nations
XE_{ijt}	Exchange rate between China and trade partner in year t	USD	IMF: International Financial Statistics
θ_j	Dummy variable for landlocked countries (1= landlocked and 0 = not landlocked)	Dummy Variable	Geography Realm

Table 2: Description and Data Sources of Variables

16 countries (out of 93) are landlocked, which are mainly from Africa and Asia. For exchange rates, data is taken from the IMF's International Financial Statistics and equation 4 is used to determine the exchange rate between China and the southern trade partners. The descriptive statistics, of each variable used in this analysis, are shown in the Table 3.

This year's data is excluded from the analysis due to the occurrence of huge degrees of variations. As the COVID pandemic hit the world economy, it impacted the trade activities, too, very harshly. The drop of 14 percent, between Africa and China, was seen in the first quarter of 2020 and it is expected to worsen in near future[10]. Similar data has been seen in the other parts, too. Since the industrial activities are halted and some economies are witnessing catastrophic implications of COVID-19 on the overall economic conditions, it will not be analytically prudent to include the data of this year.

[10] Asiedu, K,G. (2020) *Trade between China and Africa dropped 14 percent in the first quarter and could get worse*, Quartz. Available at: link (Last accessed: August 8, 2020)

	$ln(X_{ijt})$	$ln(distance_{ij})$	$ln(GDP_{it})$	$ln(GDP_{jt})$	$ln(Pop_{it})$	$ln(Pop_{jt})$	XE_{ijt}	θ_j
Total Observations	1767.00	1767.00	1767.00	1767.00	1767.00	1767.00	1767.00	1767.00
Minimum Value	5.04	7.94	27.92	18.15	20.96	11.15	0.00	0.00
Maximum Value	18.56	9.87	30.29	28.69	21.06	21.04	28.09	1.00
Range	13.51	1.92	2.37	10.54	0.09	9.88	28.08	1.00
Sum	23152.13	16306.15	51737.06	41291.49	37130.71	27595.30	3187.32	304.00
Median	13.25	9.34	29.44	23.32	21.01	16.01	0.18	0.00
Mean	13.10	9.23	29.28	23.37	21.01	15.62	1.80	0.17
SE.Mean	0.06	0.01	0.02	0.05	0.00	0.05	0.10	0.01
CI.Mean.0.95	0.11	0.02	0.04	0.09	0.00	0.09	0.19	0.02
Variance	5.83	0.16	0.64	3.70	0.00	3.97	16.58	0.14
Standard Deviation	2.42	0.39	0.80	1.92	0.03	1.99	4.07	0.38
Coefficient of Variation	0.18	0.04	0.03	0.08	0.00	0.13	2.26	2.19

Note: This table shows descriptive statistics of the data scaped by the author for performing the analysis. All the variables used under gravity model are taken under logarithmic scale, therefore descriptive statistics shown of variables are in the logarithmic scale.

Table 3: Descriptive Statistics

Results

Ordinary Least Squares (OLS)

Before applying the gravity model, results from the panel regression are also interpreted. There are three types of models used for obtaining the OLS estimates from panel data: (1) Pooled Regression, (2) Fixed Effect Model, and (3) Random Effect Model. The model of interest is a fixed effect model because it is considered as best at estimating the OLS estimates on panel data to analyze over the period. The methodology of all the three models differ. The pooled OLS model are based on the traditional OLS assumptions and treat any data as a cross-sectional data, ignoring the time and individual dimension. It can be written as:

$$\begin{aligned} ln(X_{ijt}) = \alpha + \beta_1\, ln(GDP_{it}) + \beta_2\, ln(GDP_{jt}) + \tau\, ln(D_{ij}) + \kappa_1\, ln(Pop_{it}) \\ + \kappa_2\, ln(Pop_{jt}) + \gamma_1\, \theta_j + \gamma_2\, XE_{ijt} + \epsilon_{ijt} \end{aligned} \tag{5}$$

In fixed effect model all the variables will study the fixed effects on the bilateral trade. These models are different from the traditional pooled OLS as they take difference between the individual entities into account. It can be written as:

$$\begin{aligned} ln(X_{ijt}) = \beta_1\, ln(GDP_{it}) + \beta_2\, ln(GDP_{jt}) + \tau\, ln(D_{ij}) + \kappa_1\, ln(Pop_{it}) \\ + \kappa_2\, ln(Pop_{jt}) + \gamma_1\, XE_{ijt} + \epsilon_{ijt} \end{aligned} \tag{6}$$

Similar to the fixed effects, the random effects model also take individual variations into account. Along with that it also studies time dependant variations which eliminates the biases from variables that are unobserved and change over the time. It can be written same as equation 5, although the underlying methodology differs from the pooled OLS, obviously. It can be noticed that dummy variable is dropped in this equation, this is because the variable is constant over the different data time periods and does not lead to any variation over the time.

All the variables, except the exchange rate and dummy variables, are in the natural logarithmic form. As mentioned earlier, i stands for China throughout the paper and j is the trade partners, which means $j = 1, 2, 3, \ldots, 93$, (where $j = 1, 2, 3, \ldots, 93$ are different countries in our data set). And, $t = 2001, 2002, \ldots, 2019$. Table **??** shows the results from OLS estimates under all three scenarios, for bilateral trade. The results shows the negative effect of distance between the China and trade partners, more the distance higher will be the transportation cost. In the FEM, the coefficient of distance shows the stronger negative effect on the bilateral trade, than the other two. Also, the population of the

Dependent: ln(Bilateral Trade)	Pooling Model	Fixed Effects Model	Random Effects Model
(Intercept)	295.12^{***}		273.66^{***}
	(68.53)		(69.31)
ln(distance)	-0.02	-0.78^{***}	-0.01
	(0.22)	(0.19)	(0.23)
ln(GDP Trade Partner)	0.75^{***}	0.91^{***}	0.86^{***}
	(0.10)	(0.13)	(0.09)
ln(China GDP)	1.35^{***}	1.21^{***}	1.26^{***}
	(0.16)	(0.17)	(0.16)
ln(Population Trade Partner)	0.25^{**}	0.22	0.19^{*}
	(0.09)	(0.34)	(0.08)
ln(China Population)	-16.31^{***}	-14.77^{***}	-15.24^{***}
	(3.46)	(3.70)	(3.51)
Exchange Rate	0.03		0.03
	(0.02)		(0.03)
Landlock Dummy	-1.24^{***}		-1.10^{***}
	(0.25)		(0.23)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Notes: This table shows the results of all three major types of panel regression: pooled OLS model, fixed effects model and random effects model. The best among them is considered as fixed effects model. The values in the parenthesis are the robust standard errors in the parenthesis.

Table 4: Panel Regression with Robust Standard Errors

home country has a large negative impact, under all three scenarios, on the X_{ijt}. Population can be treated as a proxy for the demand here, and due to increase in demand in the home country less amount of production is left for the foreign markets, therefore it shows a negative coefficient value for the population of China. On the contrary, the population increase for the trade partners has a little positive effect on the bilateral trade, mainly because of increase in the demand for consumption, since the majority of the part in the X_{ijt} is of exports from China. Results from PR and REM also shows the negative coefficient values for θ_j, which explains the obstacles for landlocked countries in trade activities.

Gravity Model

The results from the three different settings under the gravity models are interpreted: (a) Negative Binomial Pseudo Maximum Likelihood (NBPML), (b) Poisson Pseudo Maximum Likelihood (PPML), and (c) Gamma Pseudo Maximum Likelihood (GPML). Results from all three forms of models are in table 5. It can be inferred from the table that the results retrieved from PPML and GPML are slightly similar, but the coefficient values in NBPML are large this is because of change in the dependant variable. In NBPML, the dependant variable is not taken in the logarithmic form. The results extracted

	NBPML Model	PPML Model	GPML Model
Dependant Variable:	Bilateral Trade	ln(Bilateral Trade)	ln(Bilateral Trade)
(Intercept)	367.33^{***}	27.55^{***}	27.70^{**}
	(92.36)	(7.99)	(8.74)
ln(distance)	0.07	0.04	0.08
	(0.55)	(0.05)	(0.05)
ln(GDP Trade Partner)	0.66^{***}	0.05^{***}	0.05^{***}
	(0.02)	(0.00)	(0.00)
ln(China GDP)	1.38^{***}	0.11^{***}	0.12^{***}
	(0.17)	(0.01)	(0.02)
ln(Population Trade Partner)	0.21^{***}	0.02^{***}	0.02^{***}
	(0.02)	(0.00)	(0.00)
ln(China Population)	-19.66^{***}	-1.43^{***}	-1.44^{***}
	(4.62)	(0.40)	(0.44)
Exchange Rate	0.02^{***}	0.00^{***}	0.00^{***}
	(0.01)	(0.00)	(0.00)
Landlock Dummy	-1.45^{***}	-0.10^{***}	-0.10^{***}
	(0.07)	(0.01)	(0.01)
Log Likelihood	-25805.27		-2766.41
Deviance	2012.62	160.57	14.31
Total Observations	1767	1767	1767

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Notes: This table shows the results from the three different types of gravity model analysis: (a) Negative Binomial Pseudo Maximum Likelihood (NBPML), (b) Poisson Pseudo Maximum Likelihood (PPML), and (c) Gamma Pseudo Maximum Likelihood (GPML). In NBPML, dependant variable taken is the bilateral trade since the underlying methodology of NBPML does not allow to take the dependant variable in logarithmic scale. Remaining two, PPML and GPML, has dependant variable and independent variables, both, in the logarithmic scale.

Table 5: Gravity models analysis

from using gravity equations under three different settings are the same as interpreted in panel regression. However, the coefficient value for distance, here, is positive but very small which shows no effect of transportation cost in the bilateral trade between China and the southern countries.

Increase in the size of economy (GDP) has positive effects on the bilateral trade, as the economic mass of trade partner influences the trading activities and larger mass will imply more tendency to engage in the trade. The limitations of landlocked countries is seen in the results here, too, coefficient values under all three settings have a negative sign. However, the researchers such as Bergstrand (1985), Dell'Ariccia (1999) and De Benedictis and Taglioni (2011), have argued for adding exchange rate in the gravity equation, but, in both panel regression and gravity model analysis, here, it shows a very little effect due to exchange rate in the bilateral trade relations between China and trade partners.

Conclusion

This paper studies the factors affecting the bilateral trade between China and the southern countries. Being the largest player in the south-south trade bolsters the significance of studying its role in the trade activities. From the data set of 93 southern countries, which are engaged in trade with China in one way or another, this paper explores the effects of geographic constraints, exchange rates, population size, economic size, and distance on the bilateral trade volume between China and the global south. Using the gravity analysis, results show a minor (positive) effect of distance on the bilateral trade activities. Astonishingly, it finds no effect of exchange rates on the total trade volume. China's engagement in the south-south trade can be explained by its need for raw materials, which led to expanding the trade ties with small countries, including the island nations of Africa and Latin America and, in return, it is exporting the manufactured goods to them. In south-south trade, China accounts for more than 70 percent of market gains (Aksoy and Ng, 2013). The favorability of terms of trade in these transitions is still an open question, and, to our knowledge, no studies have been done to address these, yet. Not taking the terms of trade in account is one of the major limitations of this analysis.

Similarly like Coulibaly and Fontagné (2006), this paper also confirms the disadvantages for the landlocked countries in international trade. The landlocked countries due to geographical challenges face a lot of difficulties in engagement. Built road initiative (BRI) can be very beneficial for the these countries through connecting them with each other through a road. BRI's agenda of establishing the trade routes, connected with a single road, to Europe, will empower the landlocked countries in between to carry on the trade activities overcoming their geographical limitations. Further, I find no effect of exchange rate fluctuations on bilateral trade flows, which reflects that south-south trade cannot be disturbed by the exchange rate instability. Although the previous analysis drawn have argued that the exchange rates can be a major issue in affecting the bilateral trade volume. From the gravity models, the results showed that transportation cost is not a cause of any hindrances in the bilateral trade with the partner countries. The results also interpret a strong effect of economic mass, of both the home country and trade partners on the bilateral trade. Similarly the population growth of the trading partners has positive effects on the total volume of trade, because of increase in consumption demand subsequently but the population growth in the home country has tremendously negative effect, since the demand for consumption in the home has now increased, in compare to the early years, and resulting more consumption of goods at home.

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